

EXPERT DECISION REPLAY PLATFORM

Milestone 2 (Week 3-4) Technical Implementation & Verification Report

Project Title: Expert Decision Replay Platform | **Phase:** Milestone 2 (Week 3-4)
Deliverables Completed: Decision Management Module, Alternative Comparison Matrix, File Attachments, Discussion Threads, Version Snapshot Tracking.
Verification Status: 100% Automated Test Pass Rate (pytest) & Live Web UI Functional

1. Executive Summary

Milestone 2 expands the **Expert Decision Replay Platform** from foundational authentication and role-based access control (RBAC) into full decision management capabilities. The system allows employees, reviewers, managers, and administrators to document organizational decisions, construct structured alternative comparison matrices, collaborate via discussion threads, attach supporting documentation, and maintain immutable version snapshots of all decision revisions.

2. Milestone 2 Deliverables Summary

Module	Specification Task	Implementation Details & Endpoints	Status
Decision Management	Full Decision CRUD & Lifecycle Statuses	POST/GET/PUT/DELETE /api/v1/decisions Categories: Architecture, Infrastructure, Security, Process. Statuses: Draft, Under Review, Approved, Rejected, Archived.	COMPLETED
Alternative Analysis	Pros/Cons, Costs, Risk Profile & Composite Matrix	POST/GET /api/v1/decisions/{id}/alternatives Calculates feasibility scores, risk weights, cost penalties, and auto-recommends optimal options.	COMPLETED
Discussion Module	Collaborative Comments & Decision Rationale	POST/GET/DELETE /api/v1/decisions/{id}/comments Threaded discussion notes with user role attribution.	COMPLETED
File Management	Upload & Download Decision Documents	POST /api/v1/decisions/{id}/upload Serves local storage uploads from static/uploads/.	COMPLETED
Version Tracking	Automatic Revision Snapshots	GET /api/v1/decisions/{id}/versions Every PUT update increments version and saves an immutable snapshot in decision_versions table.	COMPLETED

3. Expanded Database Schema & ORM Models

Milestone 2 introduced 3 new database tables and updated the decisions table:

- **decisions:** Added rationale text column and relational cascades to alternatives, versions, comments, and attachments.
- **decision_versions:** Stores revision snapshots (decision_id, version, title, problem_statement, category, status, rationale, change_summary, created_by_id).
- **comments:** Threaded discussion notes (decision_id, user_id, parent_id, content, created_at).
- **attachments:** Uploaded document metadata (decision_id, uploaded_by_id, filename, file_path, file_size, content_type).

4. Automated Verification & Test Results

Both Milestone 1 and Milestone 2 test suites were executed via pytest, verifying 100% pass rates across authentication, decision CRUD, alternative evaluation, discussion comment posting, file storage, and version tracking.

```
$ pytest tests/test_milestone1.py tests/test_milestone2.py -v ===== test
session starts ===== tests/test_milestone1.py::test_health_check PASSED
[ 8%] tests/test_milestone1.py::test_user_login_admin PASSED [ 16%]
tests/test_milestone1.py::test_user_registration PASSED [ 25%]
tests/test_milestone1.py::test_protected_me_endpoint PASSED [ 33%]
tests/test_milestone2.py::test_health_check_m2 PASSED [ 41%]
tests/test_milestone2.py::test_list_decisions_seed PASSED [ 50%]
tests/test_milestone2.py::test_filter_decisions_by_category PASSED [ 58%]
tests/test_milestone2.py::test_create_decision_with_alternatives PASSED [ 66%]
tests/test_milestone2.py::test_update_decision_version_bump PASSED [ 75%]
tests/test_milestone2.py::test_alternative_comparison_matrix PASSED [ 83%]
tests/test_milestone2.py::test_discussion_comments PASSED [ 91%]
tests/test_milestone2.py::test_file_upload_and_download PASSED [100%]
===== 12 passed in 1.42s =====
```

5. Conclusion & Next Steps (Milestone 3 & 4)

Milestone 2 has been successfully completed and verified. All core decision recording, alternative evaluation, discussion, file attachment, and version tracking requirements are active and functional in the single page web interface (SPA) and FastAPI backend.

Upcoming Phase — Milestone 3 (Week 5-6):

- Multi-level approval workflows with assigned reviewer escalation.
- Real-time notification system for status updates and comment mentions.
- Organizational analytics dashboards and PDF/Excel decision exports.